Implementation of Algorithm on Intraday Trading Bot for Trading Foreign Exchange Markets Using Python

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Abstract: **The Python programming language is used to design and execute an algorithm for an intraday trading bot on the forex market. The algorithm is based on technical analysis using multiple indicators and signals and is designed to generate buy and sell signals based on market conditions. The bot is back tested and simulated on historical market data, and its performance is evaluated using various metrics. Live trading results are also presented, and the bot is compared with a benchmark to determine its effectiveness. The study also includes an analysis of risk management strategies. The implications of this research are significant for traders and investors who wish to use algorithmic trading to improve their trading outcomes. Overall, this research demonstrates the potential for algorithmic trading to generate consistent profits in intraday trading on the foreign exchange market.**

***Keywords**: Intraday trading, Algorithmic trading, Foreign exchange market, Python programming, Technical analysis, Trading bots, Back testing, Performance evaluation, Risk management, Trading strategies.*

I. INTRODUCTION

The introduction of this research paper lays the foundation for the study of the implementation of an algorithm for an intraday trading bot for trading foreign exchange markets using Python. The background section provides an overview of intraday trading and the rise of algorithmic trading, which has revolutionized the way traders and investors approach the financial markets.[1] The aim of this paper is to design and implement an algorithm for an intraday trading bot, which can be used to trade the foreign exchange market with the help of Python programming language. Algo-trading has gained popularity in recent years, enabling traders and investors to utilize sophisticated mathematical models and algorithms to make informed trading decisions in a timely and automated manner. This research paper seeks to contribute to the field of algorithmic trading by designing and implementing an intraday trading bot that can be used to trade the foreign exchange market. The objective Python and search paper are to design an algorithm for an intraday trading bot, implement the algorithm in Python, and evaluate the performance of the bot on the foreign exchange market.

A. *Background*

Explanation of intraday trading **-** The practice of purchasing and selling financial assets during the same trading day, to profit from brief fluctuations in price, is known as intraday trading. This type of trading requires a close monitoring of the markets, quick decision-making abilities, and an in-depth understanding of technical analysis and market behavior. Intraday traders often make use of real-time market data to make informed trading decisions, with the goal of taking advantage of short-term price movements in the financial markets

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The rise of algo-trading – Algo-trading enables traders and investors to utilize sophisticated mathematical models and algorithms to make automated trading decisions in the financial markets. The rise of algorithmic trading has been driven by advancements in technology and the increasing availability of market data. Algorithmic trading turnedout to be an efficient way of trading the financial assetsand has become an integral part of the modern tradinglandscape.[9]

The main aim of this research paper - The main goal is to design, implement, and assess an algorithm for an intraday trading bot, which can be used to trade the foreign exchange market with the help of Python programming language. The objective of this researchpaper is to enhance the current body of literature on algorithmic trading by creating and executing a tradingbot suitable for intraday trading in the forex market. The goal of this research paper is to provide a useful tool for traders and investors in the foreign exchange market, and to demonstrate the potential of algorithmic trading.

B. *Objectives*

To design an algorithm for an intraday trading bot - The main aim of this research paper is to developan algorithm for an intraday trading bot that can be used to trade the foreign exchange market. This will involve the selection and combination of technical analysis and trading rules that will be used to generate trading signals. The algorithm will be designed to consider various market conditions and to adapt to changing market conditions in real time.

To implement the algorithm in Python - The second objective of this research paper is to implementthe algorithm in Python, a popular programming language that is widely used for financial and quantitative analysis. The implementation of the algorithm in Python will allow for the automation of the trading process and the execution of trades in real-time, as well as providing an environment for back testing and performance assessment. [2]

To analyze the performance of the bot - The finalobjective is to analyze the performance of the intradaytrading bot on the foreign exchange market. This will involve the use of back testing and simulation methodsto assess the accuracy of the trading signals generated

by the algorithm, as well as an analysis of the performance of the bot in terms of its profitability andrisk management. The evaluation of the performance of the bot will provide valuable insights into the effectiveness of the algorithm and its potential for usein real-world trading scenarios.

II. LITERATURE REVIEW

A. *Overview of Algorithmic Trading*

Advantages and disadvantages – There are severaladvantages and disadvantages of algo-trading in financial markets. Some of the advantages of algorithmic trading include better efficiency and speedthan humans, elimination of emotions (fear & greed) in decision-making, and the capacity to manage substantial volumes of data and execute intricate computations. On the other hand, some of the disadvantages of algorithmic trading include the potential for errors or technical malfunctions, delay inorder execution, increased risk of market manipulation, and the need for a high level of technical knowledge and resources. [3]

Overview of intraday trading bots - Intraday trading bots are a type of algorithmic trading that is specifically designed for intraday trading. These bots use advanced mathematical models and algorithms to make automated trading decisions, with the goal of taking advantage of short-term price movements in thefinancial markets. Automated bots for intraday tradingcan be employed for trading an extensive variety of financial assets such as stocks, crypto, forex etc. [10]

B. *Previous Research on Intraday Trading Bots*

Comparison with other trading bots - Previous research on intraday trading bots has compared the performance of these bots with other types of trading bots, such as high-frequency trading bots and mean- reversion trading bots. These comparisons have shownthat intraday trading bots can be somewhat effective for short-term trading, but most of them do not perform as well as other types of trading bots over longer periods of time.[11]

Analysis of performance - Previous research on intraday trading bots has also analyzed theperformance of these bots

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in terms of profitability andrisk management. These analyses have shown that intraday trading bots can be profitable but may also come with a higher level of risk due to their reliance on short-term price movements.

C. *Review of Foreign Exchange Market Characteristics and behavior*

With a daily trading volume exceeding $6 trillion, the foreign exchange market is the biggest and most fluid financial market globally. The foreign exchange market is characterized by its high level of volatility and its tendency to respond quickly to economic and geopolitical events. Understanding the behavior of the foreign exchange market is critical for successful intraday trading in this market. [4]

***Trading volume and liquidity* -** The trading volume and liquidity of the foreign exchange market are one of the key elements that influence the efficiency and profitability of intraday trading in this market. The high trading volume and liquidity of the foreign exchange market provide traders with opportunities for quick and efficient execution of trade, as well as the ability to close out positions quickly in case of adverse market conditions.

***Impact of economic and geopolitical events* -** Theforeign exchange market is highly sensitive toeconomic and geopolitical events, and these events can influence the behavior of the market. It is important to understand the effects of economic and geopolitical events on the foreign exchange market forsuccessful intraday trading. We will take these eventsinto consideration and code the bot not to trade all these news.[5]

III. METHODOLOGY

Design of the Algorithm and Technical Indicators Used

Defining Range (DR): A time range in the New York equity trading session from 9:30 am until 10:30 am local time. The DR is marked by a box of around 12 candles in the 5-minute chart and includes the highand low of this range, including the wicks. The high and low of the DR can be used to determine with 88%probability the high or low of the day when combined with the closing price of the 5-minute timeframe.

Figure 1: Defining Range (DR) in a 5m chart

***Liquidity Imbalance Gap*:** A range in Price Delivery where one side of the Market Liquidity is offered and typically confirmed with a Liquidity Voidon the Lower Time Frame Charts in the same range ofPrice. Price can “gap” to create a literal vacuum of Trading thus posting an actual Price Gap.

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Figure 2: Imbalance Gap in a trade chart

Algorithm Logic and Rules **–** The proposed algorithm is primarily focused on the opening time of the NY Futures Market, precisely at 9:30 AM EST. Through our analysis of the past 5 years, we have observed an 85% likelihood that the day's high or lowwill be established between 9:30 AM EST and 10:30 AM, during which the price range is termed as Defining Range (DR). Subsequently, we will await theprice to breach either the high or low of DR. If the high of DR is breached post 10:30 AM EST,the low of the day will be validated, and the price will not descend to the low during the same day. Similarly, if the low is breached, the high of the day is confirmed. We will use the candle imbalance as our entry model. Once the bias for the day is ascertained via DR, the bot will enter the trade on retest of the Imbalance Gap. The Stop Loss market order will be placed at the low of the imbalance candle, while the Take Profit level will be based on a 2 RRR (Risk Reward Ratio). This implies that the bot will risk losing 1% of the account balance to make a 2% profit.

Data Sources and Processing **-** We've used the OANDA Data API as a data source for testing the algorithm. It is a reliable and easy-to-use platform foraccessing real-time and historical market data from theworld's major financial markets. The API offers a widerange of currency pairs, commodities, stock indices, and more. OANDA is a trusted name in the forex market, and their data API provides comprehensive and accurate market data for traders, investors, and developers alike. With the OANDA Data API, users can access live prices, historical data, and order book information, all in a standardized format that is compatible with most programming languages.

Risk Management and Optimization **-** Upon conducting a rigorous back testing of our strategy using the data obtained by OANDA from the past twoyears, it has been determined that the accuracy of our algorithm experiences a notable decline on days whenthe market is impacted by high-impact news. In fact, our findings indicate that the accuracy of ouralgorithm may plummet to as low as 44% on such days.[6]

IV. RESULT AND ANALYSIS

Back testing Results **-** We ran a back test of our algorithm. The following graph represents the ROI obtained monthly from Jan 2021 to December 2022.

Figure 3: Back testing result of last two years

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Profit and Loss - Quarterly Results of PNL of the bot.

Quarters and Years	2021	2022
Quarter 1	9	7
Quarter 2	8	9
Quarter 3	5	11
Quarters 4	3	13

Figure 4: Result of the 8 Quarters for two years,i.e., 2021 and 2022

Comparison with Benchmark

Comparing the 10 Year ROI of Our Algorithm with standard investment options available like Gold, Nifty, S&P500 and FixedDeposit.

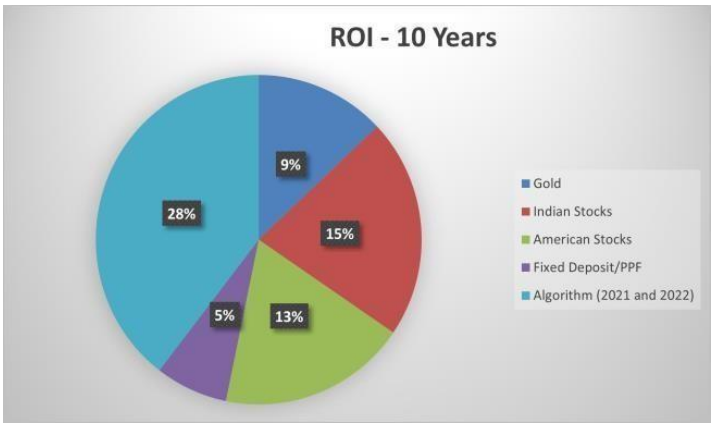

Figure 5: Comparison of ROIs of different investments

Performance and Profitability

Average ROI - 28%

DR Accuracy on Days with High Impact News - 44%

DR Accuracy on Days without High impact News -88%

V. CONCLUSION

Based on the research conducted over the past two years, it has been concluded that our algorithm exhibits promising results on major forex pairs such as EUR/USD and GBP/USD as well as American stock indices like SPX and NASDAQ. The average accuracy of Defining Range on these pairs is well above 80%. Furthermore, the average return on investment (ROI) generated by the algorithm is significantly better than that of other benchmark investments such as gold and fixed deposits. Currently, the algorithm has been deployed with demo funds and is ready to be tested in live market conditions.

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VI. IMPLICATION FOR PRACTICE

The findings of this research have significant implications for practice, particularly for intraday traders and investors who are interested in implementing algorithmic trading in the forex market. The design and implementation of the intraday trading bot, as well as its performance evaluation, have provided valuable insights into the potential benefits and limitations of algorithmic trading in this market.

Impact on Intraday Trading- The results of this research indicate that algorithmic trading can have a significant impact on intraday trading, providing traders and investors with an efficient and automated approach to making trades based on market data and technical indicators. The use of algorithmic trading can help traders and investors make more informed and strategic decisions, as well as minimize the risk ofhuman error in trading. [7]

Significance for Traders and Investors- The results of this research also have implications for traders and investors who are interested in optimizingtheir trading strategies and improving their performance in the foreign exchange market. The use of algo-trading, as demonstrated in this research, can help traders and investors achieve their investment objectives and make more profitable trades.[8]

In conclusion, the results of this research highlightthe significance of algorithmic trading in intraday trading and the foreign exchange market. The design and implementation of the intraday trading bot, as well as its performance evaluation, have provided valuable insights into the potential benefits and limitations of algorithmic trading in this market. These findings have important implications for traders and investors who are interested in optimizing their trading strategies and improving their performance in the foreign exchange market.

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